Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/24/2021
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	Plant RNA Regulation Redux (in MVP--Multi-use Variable-g Platform)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	05/31/2019	**End Date:**	05/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/30/2022 per NSSC (Ed., 3/2/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1212		
Performance Goal No.:			
Performance Goal Text:			
	The spaceflight experiment Plant RNA Regulation (PRR) was designed to generate comprehensive datasets for gene expression, small RNA species, and protein profiles from wild type Arabidopsis grown under the same set of physiologically relevant experimental conditions. Given the cost and limited availability of spaceflight opportunities, the goal was to maximize the scientific potential of the experiment. The experiment utilized the European Modular Cultivation System (EMCS) facility, which provides two critically important parameters for plant spaceflight experiments (Vandenbrink and Kiss, 2016). These are the capacity for providing a “1 g control in space” and a lighted environment for plant growth. PRR had two major objectives: To profile the transcriptional and post-transcriptional changes associated with early shoot and root development, and to monitor and characterize protein profiles of developing shoots and roots.		

Task Description:	Unfortunately, due to unknown factor(s), germination of PRR was reduced and growth of the seedlings was compromised. Based on poor growth and variability in growth, there was insufficient tissue to generate RNA for both RNA-Seq and small RNA sequencing for root and shoot tissue for all the samples. Additionally, there was insufficient tissue for the proteomic work. By careful examination of seedlings in the cassettes, we have been able to harvest individual seedlings from select seed cassettes that exhibited nominal growth and recover shoot and root tissue. A subset of these recovered shoot samples was processed using a modified RNA isolation protocol which enabled recovery of both RNA and sRNA species from the same sample. The RNA and sRNAs were subject to NextGen sequencing and preliminary analysis of these results has identified miRNAs differentially expressed between the 1g and micro g conditions. With the decommissioning of the European Modular Cultivation System (EMCS) it is not possible to have a reflight and repeat PRR in its original format. However we are confident that with some modifications and testing PRR can be adapted to fly in the newly developed Multi-use Variable-g Platform (Techshot) which would preserve two major criteria of PRR, namely the “1g control” in space and a lighted plant environment. This experiment will serve two important goals: 1. The opportunity to accomplish the major objectives of PRR (with some modifications). 2. A test of the Multi-use Variable-g Platform for plant experiments. Reference: Vandenbrink J and Kiss JZ, 2016. Space, the final frontier: A critical review of recent experiments performed in microgravity. Plant Sci 243: 115-119.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This knowledge will provide a framework for designing and improving plants that are better adapted to adverse environments, which has potential benefit both for long duration space exploration as well as on Earth in the face of global climate change. In addition, this work will contribute towards enhancing education by providing ‘hands on’ training to undergraduate students in STEM (Science, Technology, Engineering and Mathematics) programs.
Task Progress:	Progress to Date (March 2021): 1. The Principal Investigator (PI) team, Techshot, and Kennedy Space Center (KSC) have weekly/biweekly meetings to discuss testing progress, timelines, and troubleshooting of problems. The Experiment Requirements Document (ERD) was completed and the SVTRR (Science Verification Test Readiness Report) review was held in December 2020. 2. The PI team has carried out several in-lab tests using the test boxes and custom rectangular plates provided by Techshot. The following plant growth parameters, such as seed lots, plating density, duration of growth were tested and established. We have determined that two layers of Capmat provided adequate hydration to support 10-12 days of plant growth and finalized the optimal lighting and temperature conditions. 3. In order to better support plant growth, several changes were implemented by Techshot to the custom plate design. First, the height of the dish was increased to provide more growing room. Second, in order to facilitate a dry seed build with onboard hydration (just prior to the start of the flight experiment in the phytofuge module), the plates were re-fitted with a hydration port. Additionally, a plate inset was designed to keep the stack of Capmat and membrane in contact and in place. The PI team has tested plant growth with the new plate set up. 4. The PI submitted an abstract and was selected to present a pre-recorded talk in the Plant session of the 2020 Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), which was held virtually on 5-6 Nov, 2020. The talk described the test builds and plant growth studies carried out date. 5. Pre-SVT (Science Verification Test) testing in the Phytofuge module is currently underway. Seeded membranes are prepared and plates are assembled according to the finalized protocol in the PI’s lab. The dry plates are to the Techshot facility, where membranes are hydrated via the hydration port prior to the start of the experiment. We are currently establishing the growth conditions within the Phytofuge module, including temperature, lighting, and air circulation as well as testing the camera within each module for image acquisition.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Abstracts for Journals and Proceedings	Land E, Reed D, Fite J, Perera IY. "Plant Growth and Pre-Flight Testing in the MVP Phytofuge." Presented virtually at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020